

HAT-P-12b

R-0703

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_250527 · created 2026-07-27T21:59:29+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460822.8151 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.013
Transit depth (Rp/Rs)^2	2.58 +/- 0.41 [%]
Orbital Inclination (inc)	87.69 +/- 0.72
Airmass coefficient 1 (a1)	1.003 +/- 0.015
Airmass coefficient 2 (a2)	0.0 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#1 - [252, 348]
Optimal Method	PSF photometry
Transit Duration (day)	0.0909 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.161 ± 0.013 vs 0.1406 (deviation 1.37σ)
Duration fit vs published	0.0909 d vs 0.0975 d (deviation 0.95σ)
Mid-time O–C	-7.8 min
Depth SNR	10.8
Residual scatter	1.51 %

Light curve

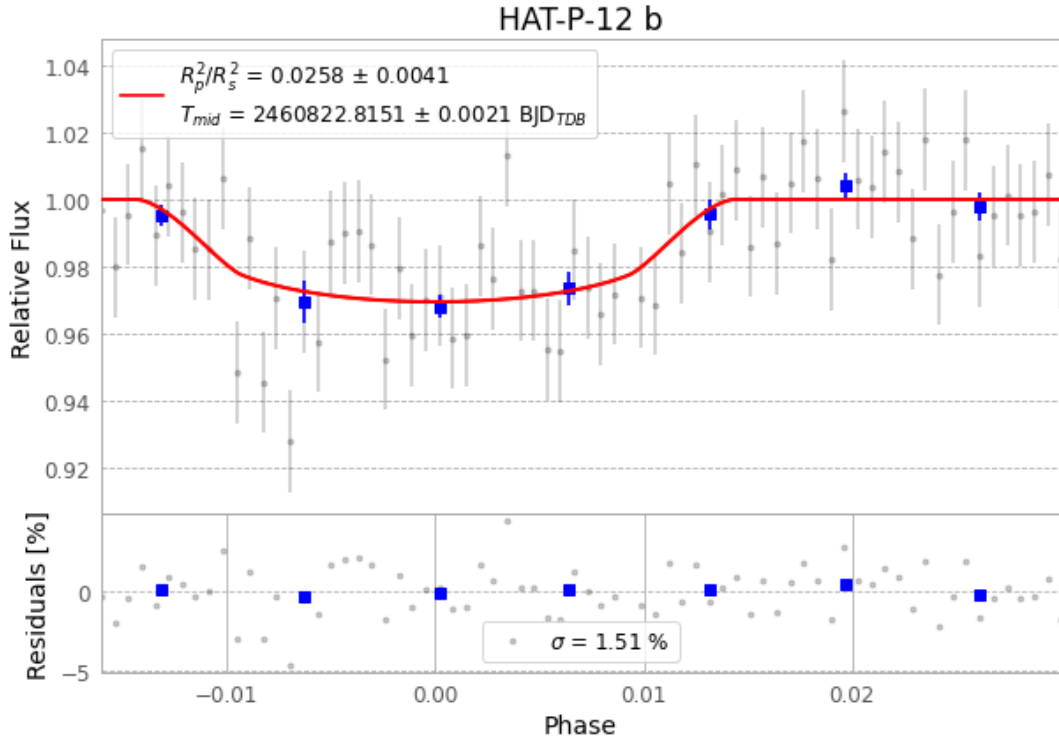


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-05-26.png (SHA-256 97001a85098e7157...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 440fddb8172547be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

72 FITS frames (47 MB), HATP-12250527061515.FITS → HATP-12250527094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250527.log (20b651a64325...), FinalLightCurve_HAT-P-12 b_2025-05-26.png (97001a85098e...), FinalLightCurve_HAT-P-12 b_2025-05-26.csv (7ced57f84817...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:59Z.